CS 4375 Assignment 3 Part 1

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1.

Prove $E_{agg} = \frac{1}{M} E_{avg}$

$$E_{agg} = E\left[\left(\frac{1}{M}\sum_{i=1}^{M} \epsilon_i(x)\right)^2\right]$$

$$= \frac{1}{M^2} E\left[\underbrace{\left(\sum_{i=1}^{N} \epsilon_i(x)\right)^2}\right]$$

$$= \frac{1}{M^2} E\left[\sum_{i=1}^{M}(\epsilon_i(x)^2) + \sum_{i=1}^{M}\sum_{j\neq i}^{M} \epsilon_i(x)\epsilon_j(x)\right]$$

$$= \frac{1}{M^2}\left[\sum_{i=1}^{M} E(\epsilon_i(x)^2) + \sum_{i=1}^{M}\sum_{j\neq i}^{M} E(\epsilon_i(x)\epsilon_j(x))\right]$$

- Note that Assumption 2 states $E(\epsilon_i(x)\epsilon_j(x)) = 0$ for $i \neq j$

$$E_{agg} = \frac{1}{M^2}\left[\sum_{i=1}^{M} E(\epsilon_i(x)^2)\right]$$

$$E_{avg} = \frac{1}{M}\sum_{i=1}^{M} E(\epsilon_i(x)^2)$$

$$E_{agg} = \frac{1}{M}\cdot\frac{1}{M}\left[\sum_{i=1}^{M} E(\epsilon_i(x)^2)\right]$$

$$\boxed{E_{agg} = \frac{1}{M} E_{avg}}$$

2. Prove $E_{agg} \leq E_{avg}$ using Jensen's Inequality $\left(f\left(\sum_{i=1}^{M}\lambda_i x_i\right) \leq \sum_{i=1}^{M}\lambda_i f(x_i)\right)$

$$E_{agg} = E\left[\left(\frac{1}{M}\sum_{i=1}^{M}\epsilon_i(x)\right)^2\right]$$

using Jensen's inequality: $E\left[\left(\frac{1}{M}\sum_{i=1}^{M}\epsilon_i(x)\right)^2\right]$

arbitrary variables

$$E_{agg} \leq \frac{1}{M}\sum_{i=1}^{M} E(\epsilon_i(x)^2), \text{ where } f = E \text{ and } \lambda = \frac{1}{M}, \; x_i = \epsilon_i(x)^2$$

As stated: $E_{avg} = \frac{1}{M}\sum_{i=1}^{M} E(\epsilon_i(x)^2)$

$$\therefore \boxed{E_{agg} \leq E_{avg}}$$

3. Prove overall training error of Adaboost is bounded by $\exp\left(-2\sum_{i=1}^{T}\gamma_t^2\right)$

Given: $D_{t+1} = \frac{D_t(i)}{Z_t} \times e^{-\alpha_t h_t(i) y(i)}$ and $H(x) = \text{sign}\left(\sum_{t=1}^{T}\alpha_j h_t(x)\right)$, we can write:

$$D_{T+1}(i) = \frac{1}{N}\cdot\frac{e^{-y_i\sum_{i=1}^{T}\alpha_t h_t(i)}}{\prod_{t=1}^{T} Z_t}$$

, where D_{T+1} is the total number of steps from 1 to T+1, hence why the initial weight and product of normalization factor Z is needed

$$D_{T+1}(i)\cdot\prod_{t=1}^{T} Z_t = \frac{1}{N}\cdot e^{-y_i\sum_{i=1}^{T}\alpha_t h_t(i)}$$

- The training error total would be $\leq \sum_{i=1}^{N} D_{T+1}(i)\prod_{t=1}^{T} Z_t$, and since weight distributions D add up to 1:

$$\boxed{Err_{train} \leq \prod_{t=1}^{T} Z_t}$$

- Z_t is the sum of weights by $Z_t = \sum_{i=1}^{N} D_t(i) e^{-\alpha_t h_t(i) y_t}$, or $\underbrace{\varepsilon_t e^{\alpha_t}}_{\text{error of incorrect classification points}} + \underbrace{(1-\varepsilon_t)e^{-\alpha_t}}_{\text{nonerror of correct classification points}}$

$$\alpha = \frac{1}{2}\ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right), \quad Z_t = \varepsilon_t e^{\frac{1}{2}\ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)} + (1-\varepsilon_t)e^{-\frac{1}{2}\ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)} = \varepsilon_t\left(\sqrt{\frac{1-\varepsilon_t}{\varepsilon_t}}\right) + (1-\varepsilon_t)\left(\sqrt{\frac{\varepsilon_t}{1-\varepsilon_t}}\right)$$

- $Z_t = \sqrt{1-\varepsilon_t}\cdot\sqrt{\varepsilon_t} + \sqrt{\varepsilon_t}\sqrt{1-\varepsilon_t} = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$
- Given $\varepsilon_t = \frac{1}{2} - \gamma_t$ per step as the total error, substitute in the above equation

$$Z_t = 2\sqrt{\left(\frac{1}{2}-\gamma_t\right)\left(1-\left(\frac{1}{2}-\gamma_t\right)\right)} = 2\sqrt{\left(\frac{1}{2}-\gamma_t\right)\left(\frac{1}{2}+\gamma_t\right)} = 2\sqrt{\frac{1}{4}-\gamma_t^2}$$

Use $1-x \leq e^{-x}$ to get $Z_t = \exp(-2\gamma_t^2)$

Substitute Z_t in Err_{train}, we get: $Err_{train} \leq \prod_{t=1}^{T} e^{-2\gamma_t^2} \rightarrow \boxed{Err_{train} \leq \exp\left(-2\sum_{t=1}^{T}\gamma_t^2\right)}$